

LECTURE # 10

COLOR IMAGE PROCESSING-II

CONTENT

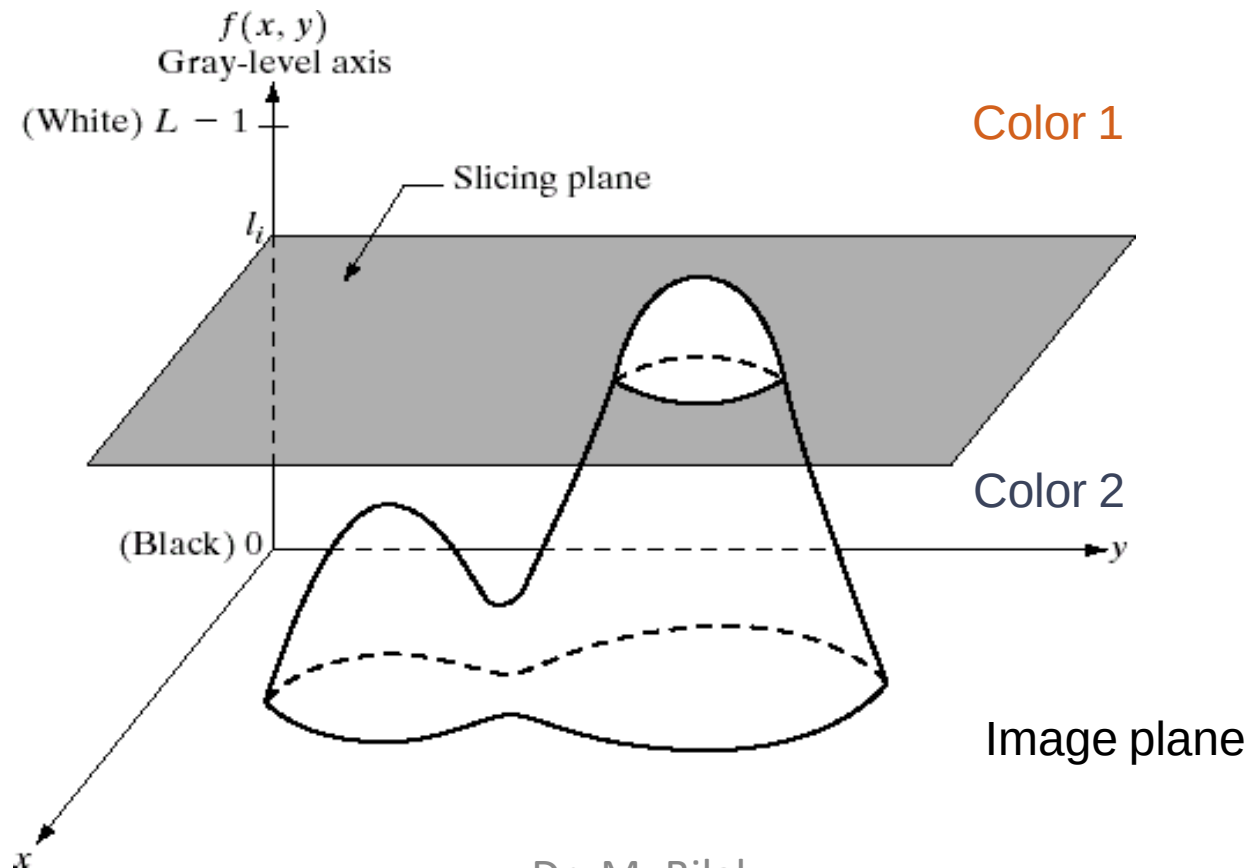
- Pseudo-color Image Processing
- Basics of Full-Color Image Processing
- Color Transformations
- Smoothing and Sharpening
- Color Segmentation
- Noise in Color Images
- Color Image Compression

PSEUDO-COLOR IMAGE PROCESSING

- Assign colors to gray values based on a specified criterion
- Differentiate the process of assigning colors to monochrome images
- For human visualization and interpretation of gray-scale events
 - Intensity slicing
 - Gray level to color transformations

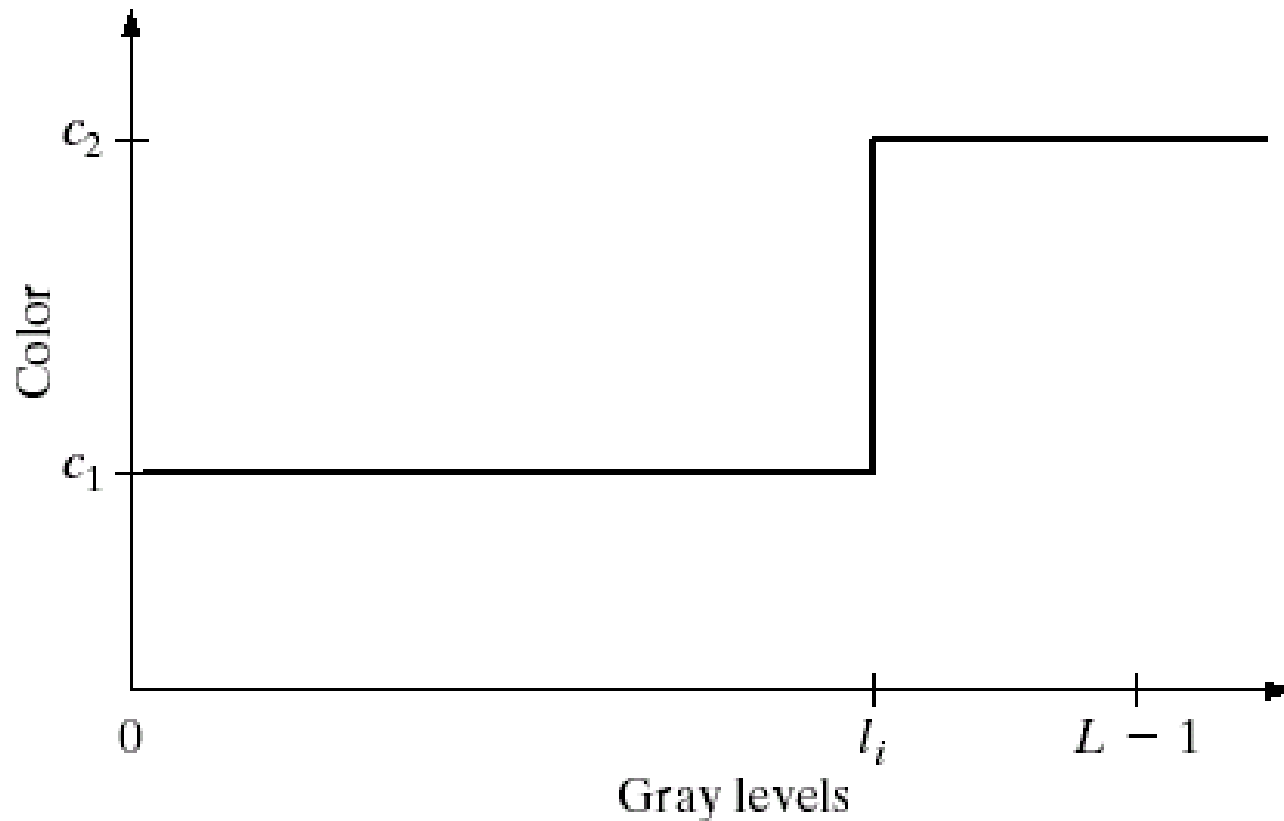
INTENSITY SLICING

- 3-D view of intensity image



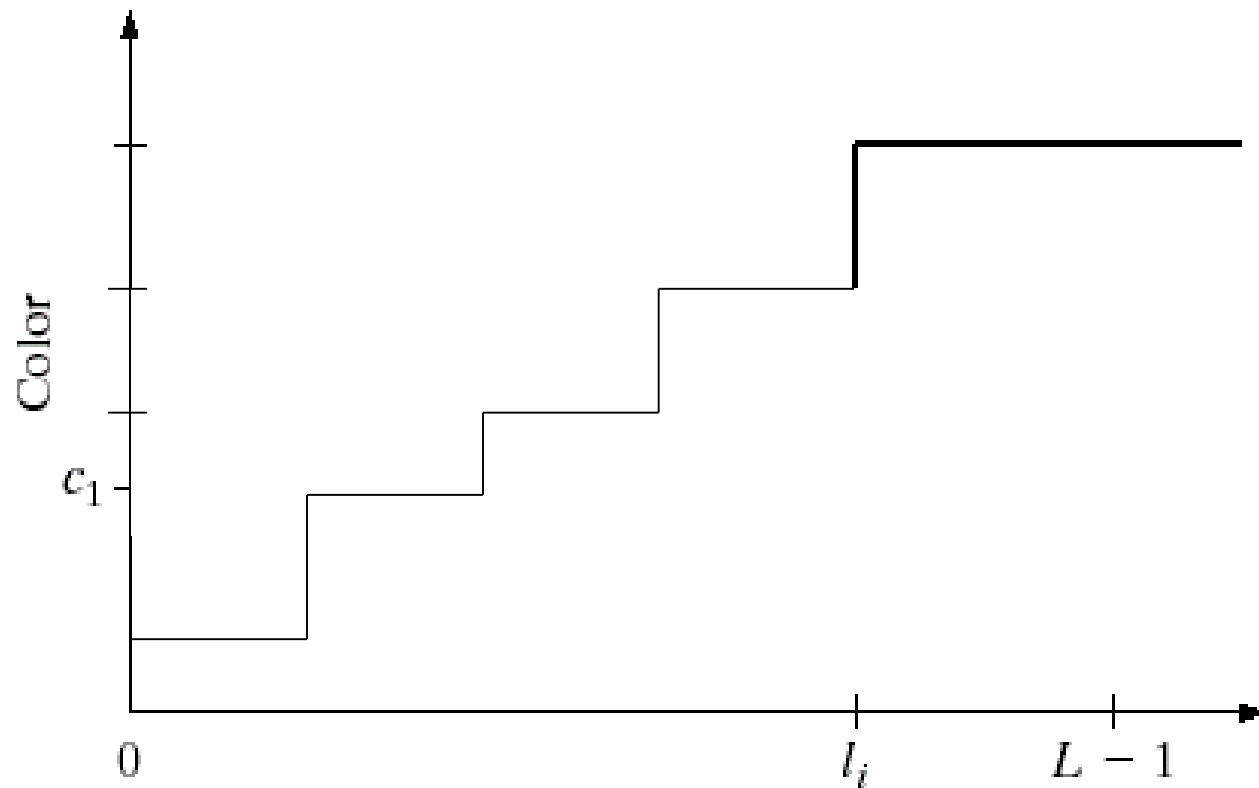
INTENSITY SLICING (CONT'D)

- Alternative representation of intensity slicing

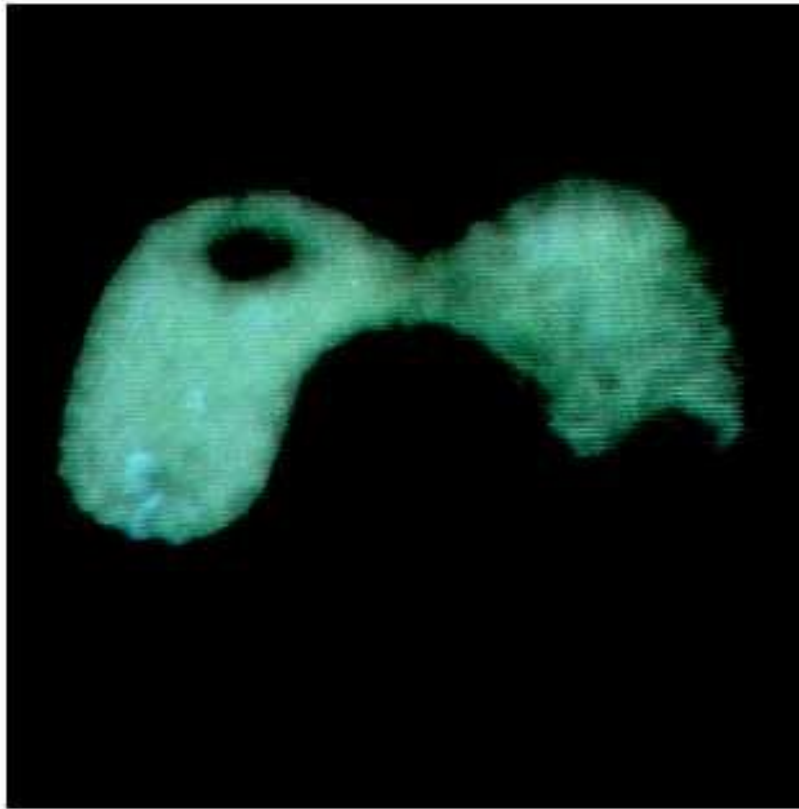


INTENSITY SLICING (CONT.)

- More slicing plane, more colors



APPLICATION 1



Radiation test pattern

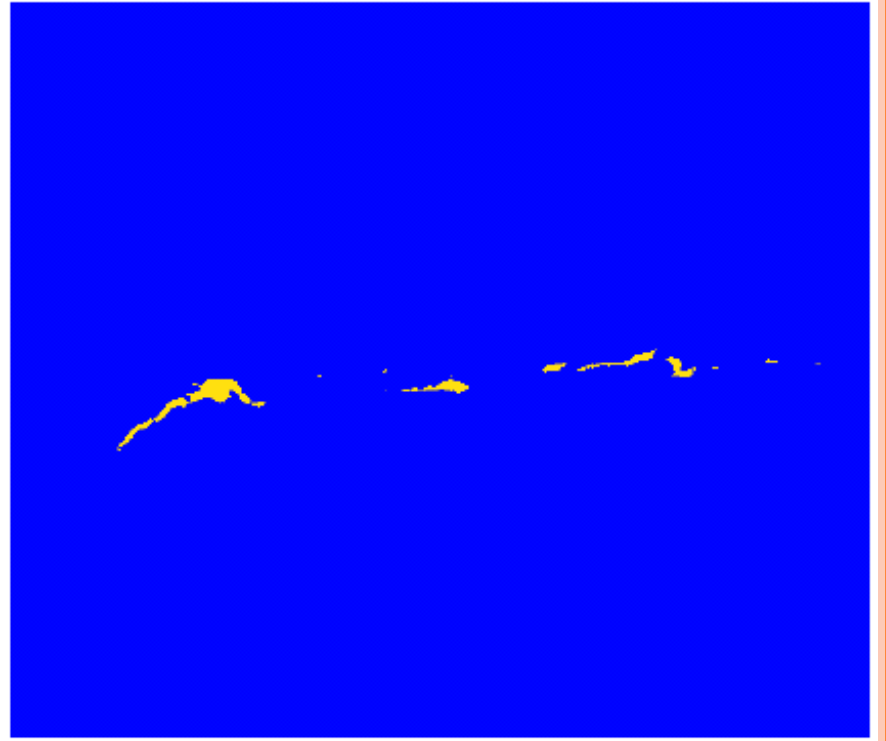
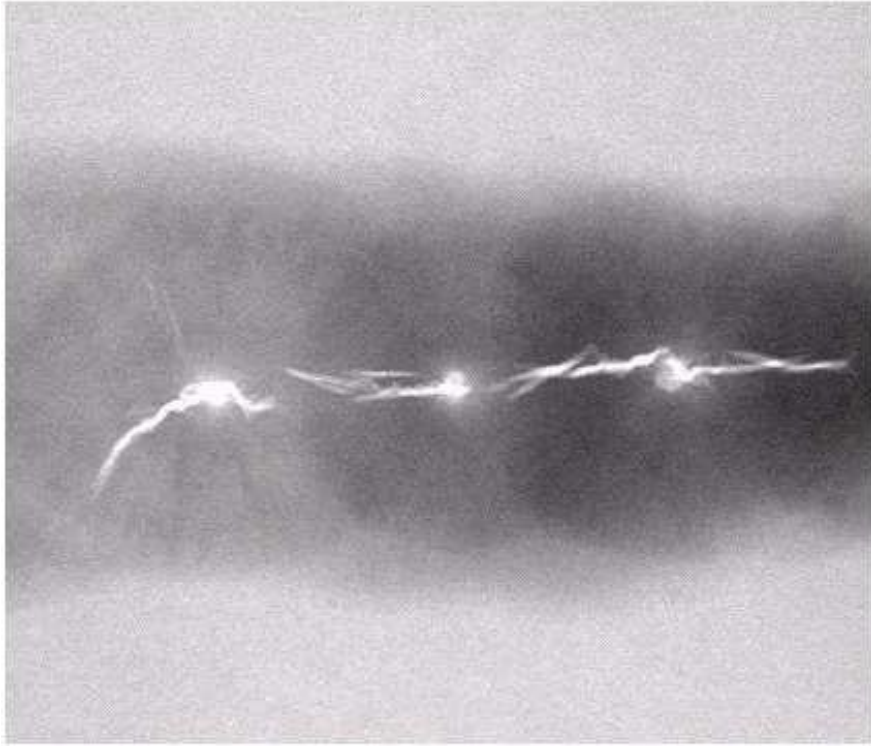


8 color regions

→
* See the gradual gray-level changes

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APPLICATION 2

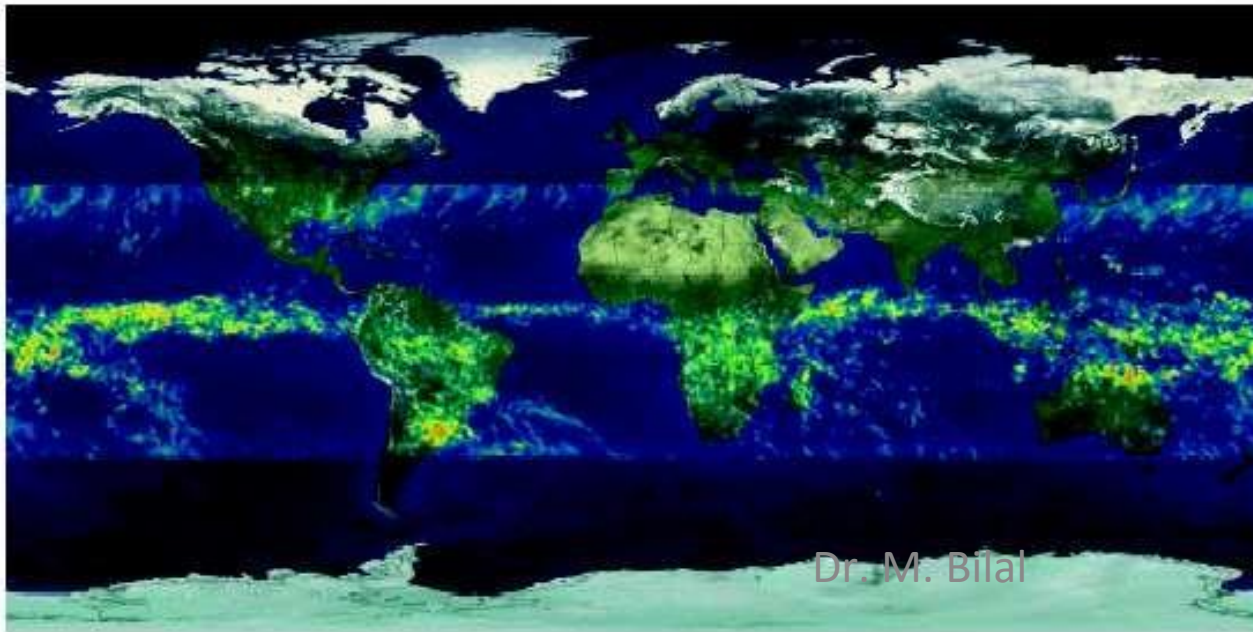
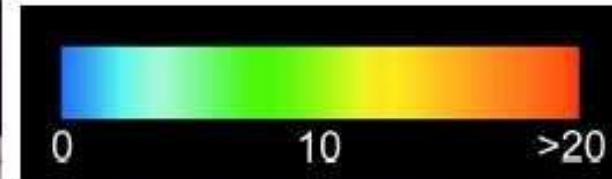


X-ray image of a weld

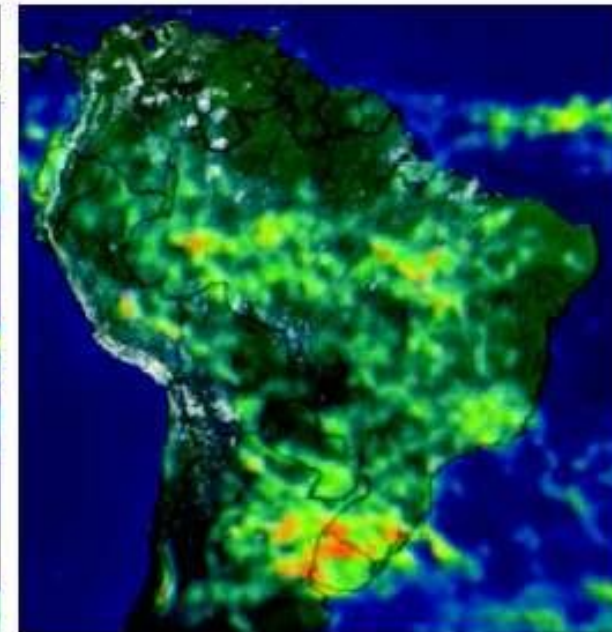
APPLICATION 3



Rainfall statistics



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GRAY LEVEL TO COLOR TRANSFORMATION

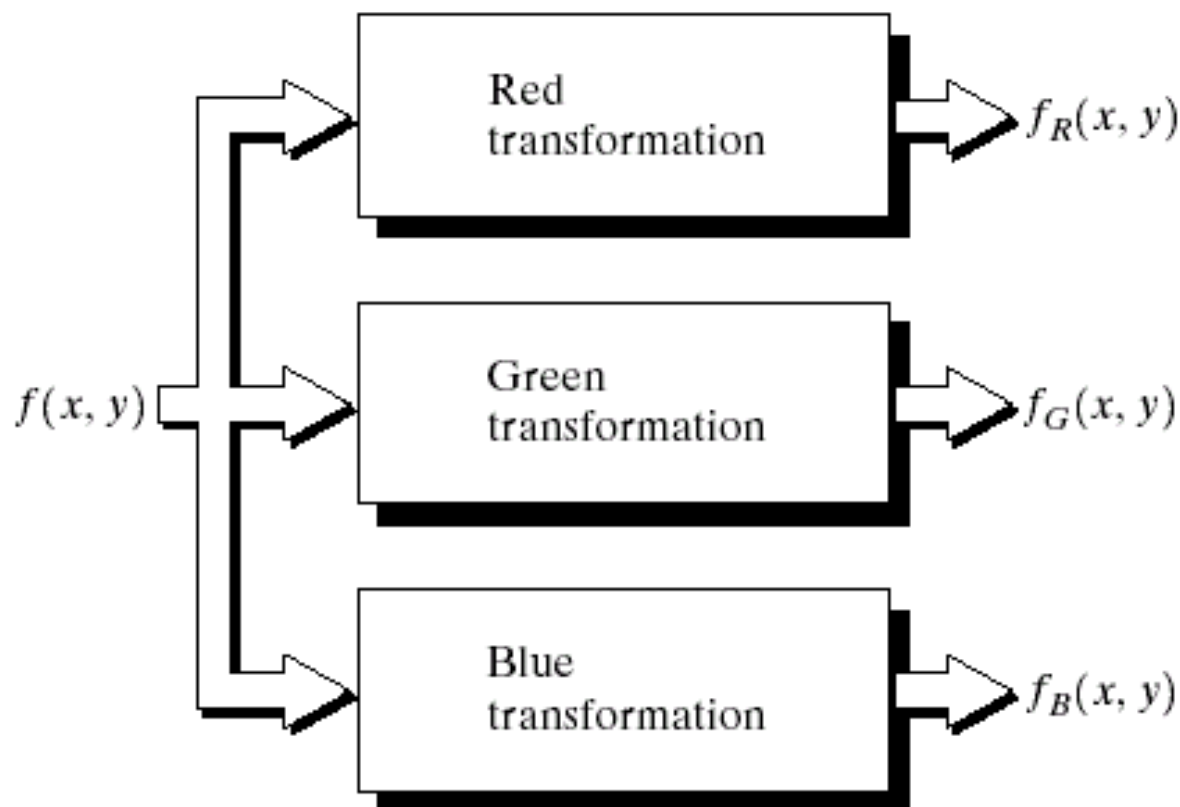
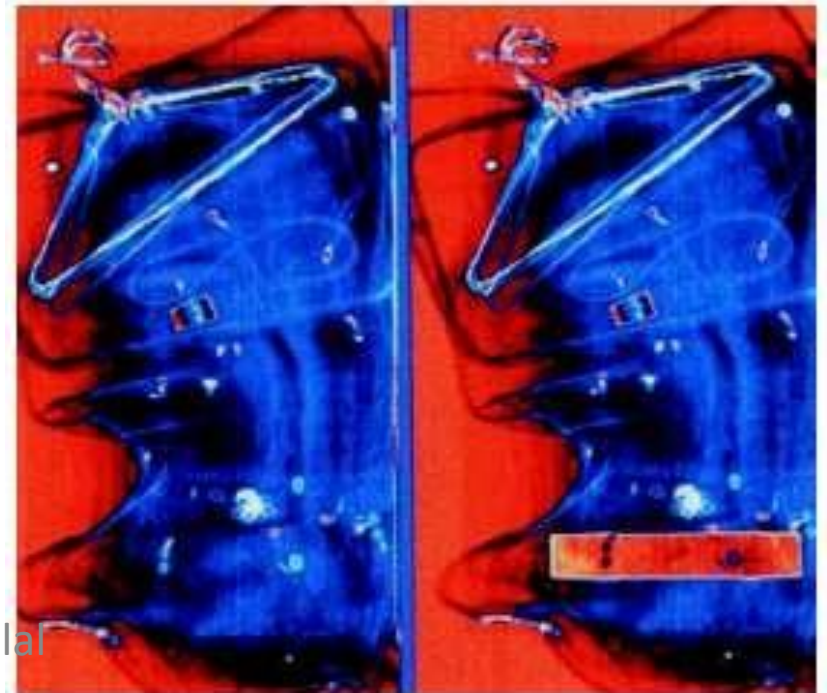
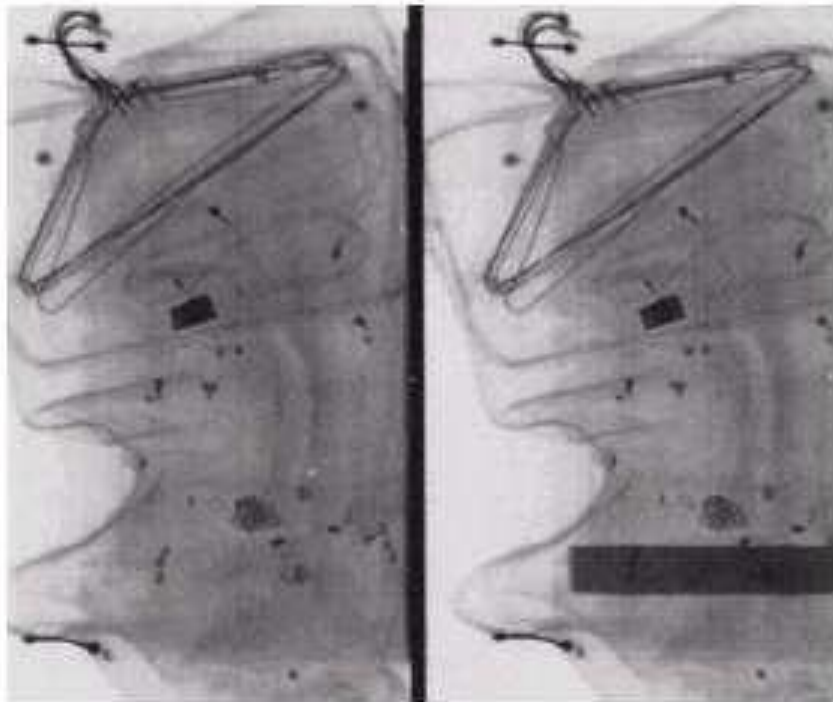
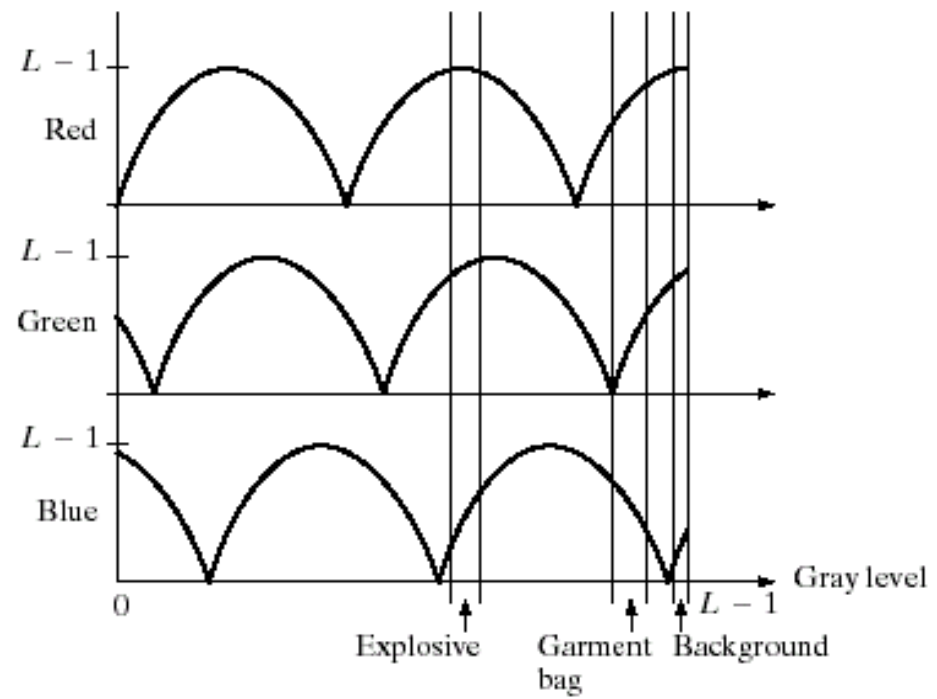


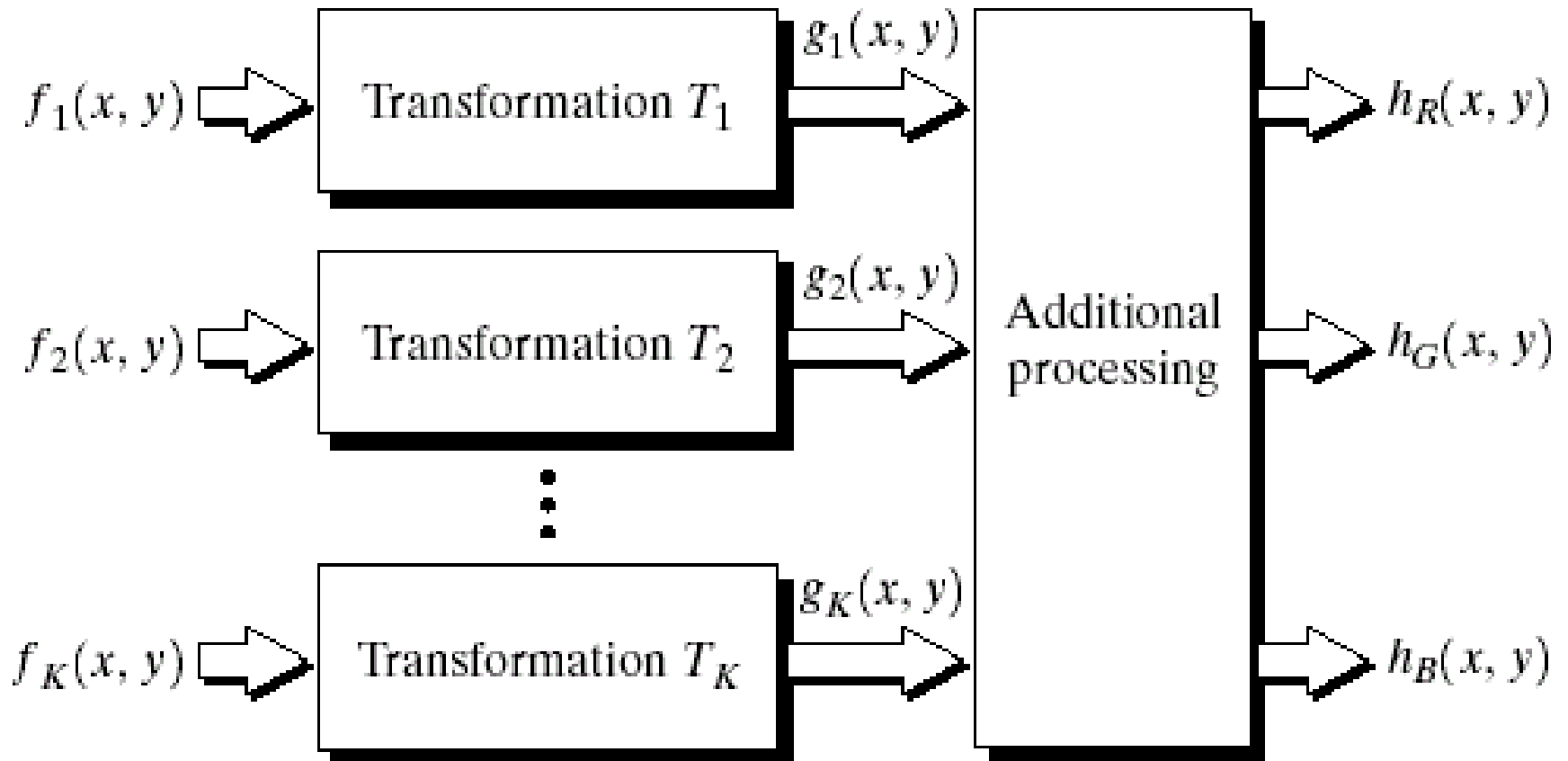
FIGURE 6.23 Functional block diagram for pseudocolor image processing. f_R , f_G , and f_B are fed into the corresponding red, green, and blue inputs of an RGB color monitor.

APPLICATION 1



COMBINE SEVERAL MONOCHROME IMAGES

Example: multi-spectral images



R



B



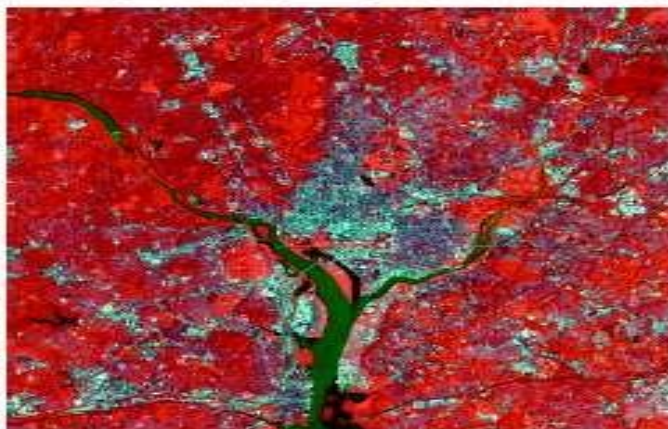
G



Near
Infrared
(sensitive
to biomass)



R+G+B



near-infrared+G+B



BASICS OF FULL COLOR IMAGE PROCESSING

- Full color image processing fall into 2 categories.
 - In 1st category we process each component image individually and then form a composite processed color image from the individually processed component.
 - In 2nd category we work with color pixels directly. Because full color images have at least three components, color pixels are really vectors.
 - Let c represent an arbitrary vector in RGB color space:

$$C = \begin{bmatrix} C_r \\ C_g \\ C_b \end{bmatrix} = \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

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BASICS OF FULL COLOR IMAGE PROCESSING

- Color components are the function of co-ordinates(x,y) so we can write it as:

$$C(x,y) = \begin{bmatrix} C_R(x,y) \\ C_G(x,y) \\ C_B(x,y) \end{bmatrix} = \begin{bmatrix} R(x,y) \\ G(x,y) \\ B(x,y) \end{bmatrix}$$

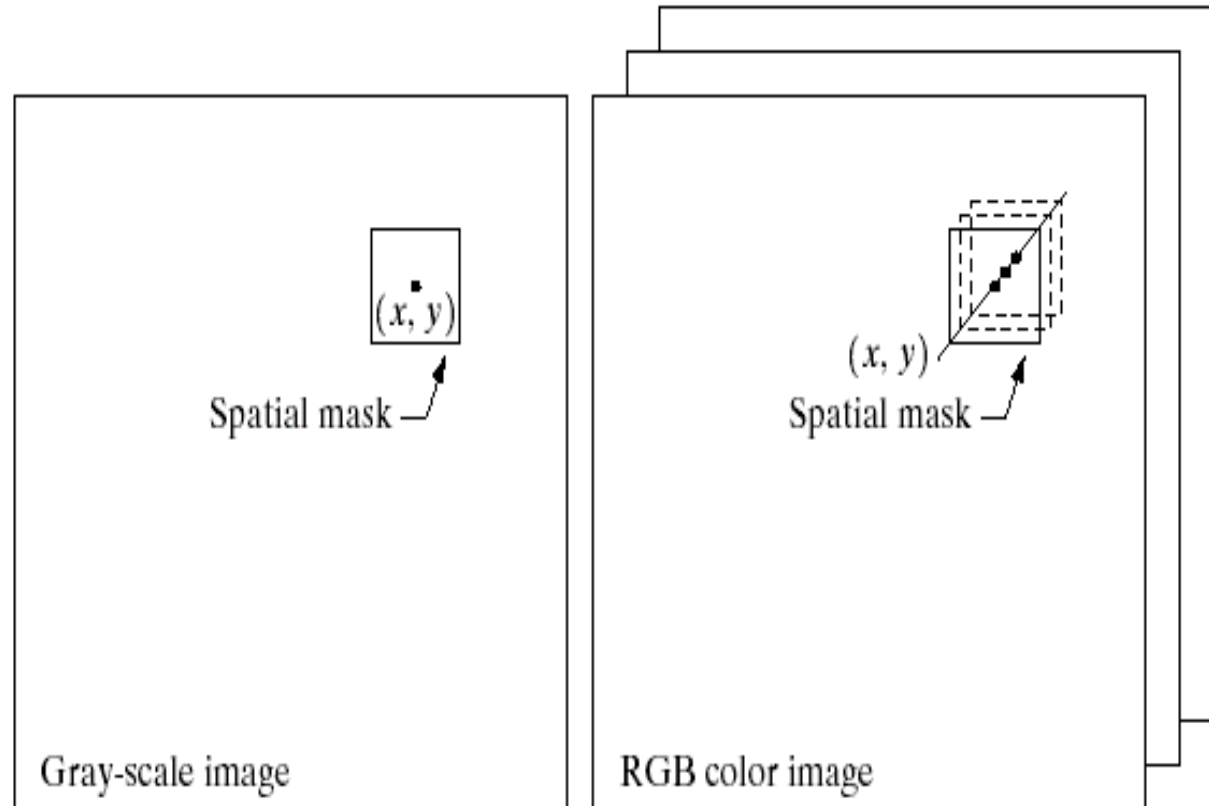
- For an image of size M x N there are MN such vectors, $c(x, y)$, for $x=0,1,2,\dots,M-1$; $y=0,1,2,\dots,N-1$

Basics of Full Color Image Processing

a b

FIGURE 6.29

Spatial masks for gray-scale and RGB color images.



COLOR TRANSFORMATIONS

- Color transformation can be represented by the expression ::

$$g(x, y) = T [f(x, y)]$$

$f(x, y)$: input image

$g(x, y)$: processed (output) image

$T[*]$: an operator on f defined over neighborhood of (x, y) .

The pixel values here are triplets or quartets (i. e group of 3 or 4 values)

COLOR TRANSFORMATIONS

- The color space chosen determine the value of n .
- If RGB color space is selected then $n=3$ & r_1, r_2, r_3 denotes the red, blue and green components of the image.
- If CMYK color space is selected then $n=4$ & r_1, r_2, r_3, r_4 denotes the cyan, hue, magenta and black components of the image.
- Suppose we want to modify the intensity of the given image using $g(x,y)=k*f(x,y)$ where $0 < k < 1$



Full color



Cyan



Magenta



Yellow



Black



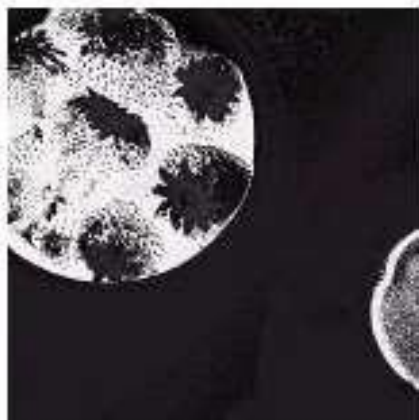
Red



Green



Blue



Hue



Saturation



Intensity



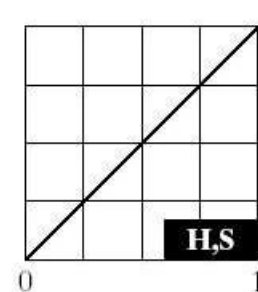
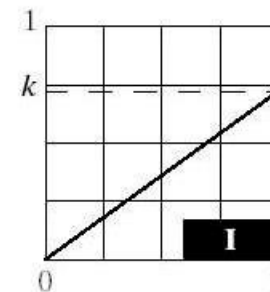
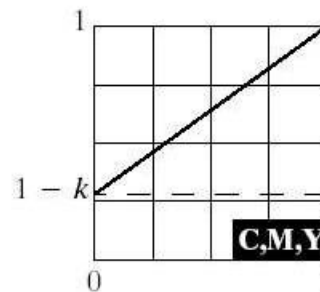
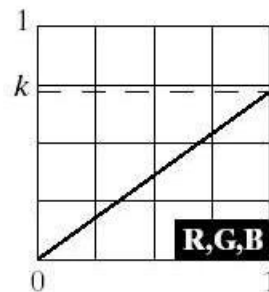
Full color

COLOR TRANSFORMATIONS

a b
c d e

FIGURE 6.31

Adjusting the intensity of an image using color transformations. (a) Original image. (b) Result of decreasing its intensity by 30% (i.e., letting $k = 0.7$). (c)–(e) The required RGB, CMY, and HSI transformation functions. (Original image courtesy of MedData Interactive.)



COLOR COMPLEMENTS

- The hues opposite to one another on the Color Circle are called Complements.
- Color Complement transformation is equivalent to image negative in Grayscale images

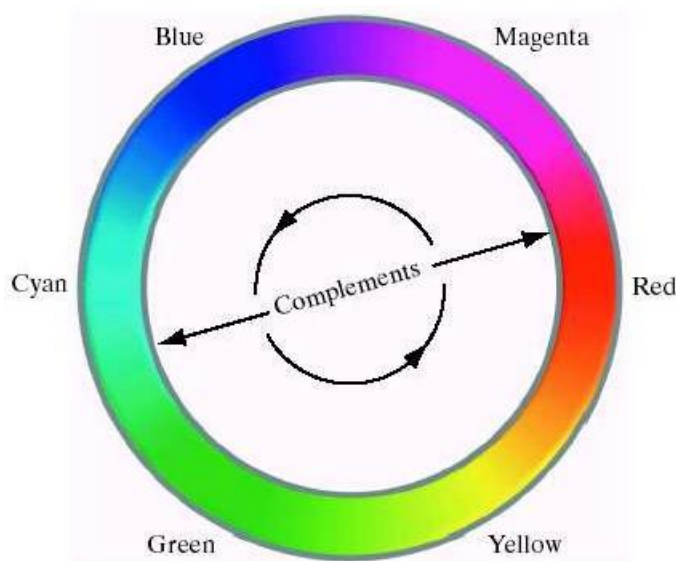
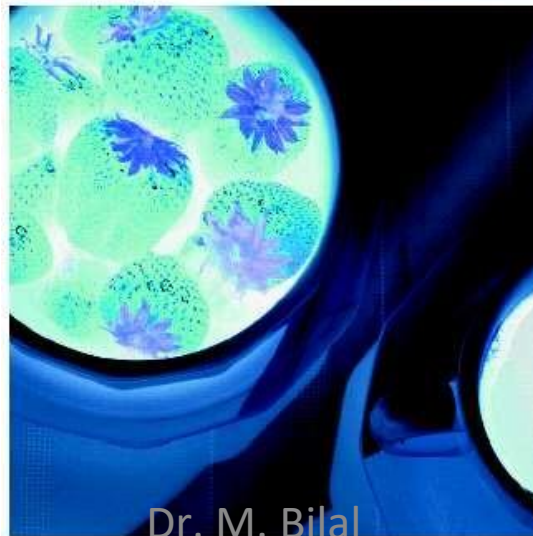


FIGURE 6.32
Complements on
the color circle.

COLOR COMPLEMENTS

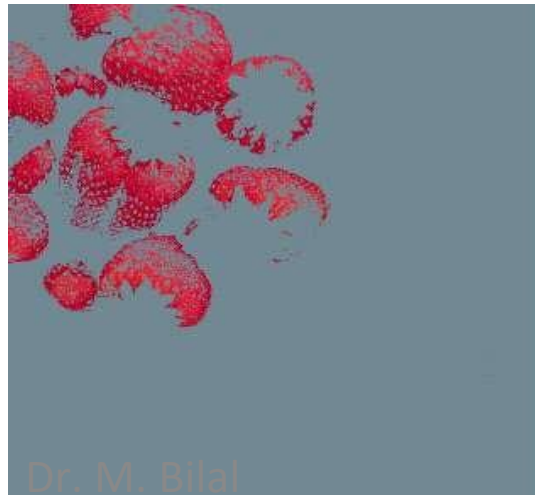


COLOR SLICING

- Highlighting a specific range of colors in an image is useful for separating objects from their surroundings.
- Display the colors of interest so that they are distinguished from background.
- One way to slice a color image is to map the color outside some range of interest to a non prominent neutral color.



Full color



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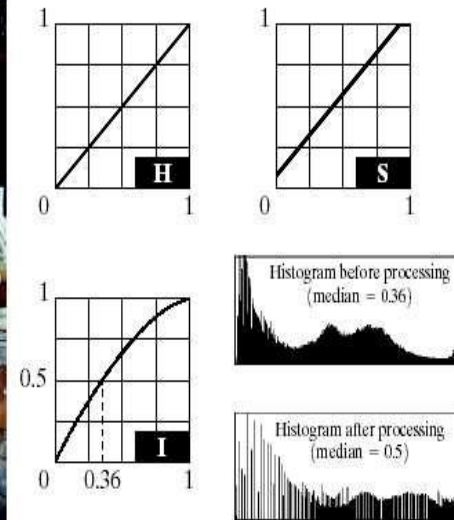


HISTOGRAM PROCESSING

- Color images are composed of multiple components, however it is not suitable to process each plane independently in case of histogram equalization. This results in erroneous color.

- A more logical approach is to spread the color intensities uniformly, leaving the colors themselves (hue, saturation) unchanged.

- HSI approach is ideally suited to this type of approach.



a b
c d

FIGURE 6.37
Histogram equalization (followed by saturation adjustment) in the HSI color space.

COLOR IMAGE SMOOTHING

- Color images can be smoothed in the same way as gray scale images, the difference is that instead of scalar gray level values we must deal with component vectors of the following form:

$$C(x,y) = \begin{bmatrix} C_R(x,y) \\ C_G(x,y) \\ C_B(x,y) \end{bmatrix} = \begin{bmatrix} R(x,y) \\ G(x,y) \\ B(x,y) \end{bmatrix}$$

- The average of the RGB component vector in this neighborhood is:

$$\bar{C}(x,y) = \frac{1}{k} \sum_{(x,y) \in S_{xy}} C(x,y)$$

COLOR IMAGE SMOOTHING

$$\bar{C}(x,y) = \begin{bmatrix} \frac{1}{k} \sum_{(x,y) \in S_{xy}} R(x,y) \\ \frac{1}{k} \sum_{(x,y) \in S_{xy}} G(x,y) \\ \frac{1}{k} \sum_{(x,y) \in S_{xy}} B(x,y) \end{bmatrix}$$

- We recognize the components of this vector as the scalar images that would be obtained by independently smoothing each plane of the starting RGB image using conventional gray scale neighborhood processing.

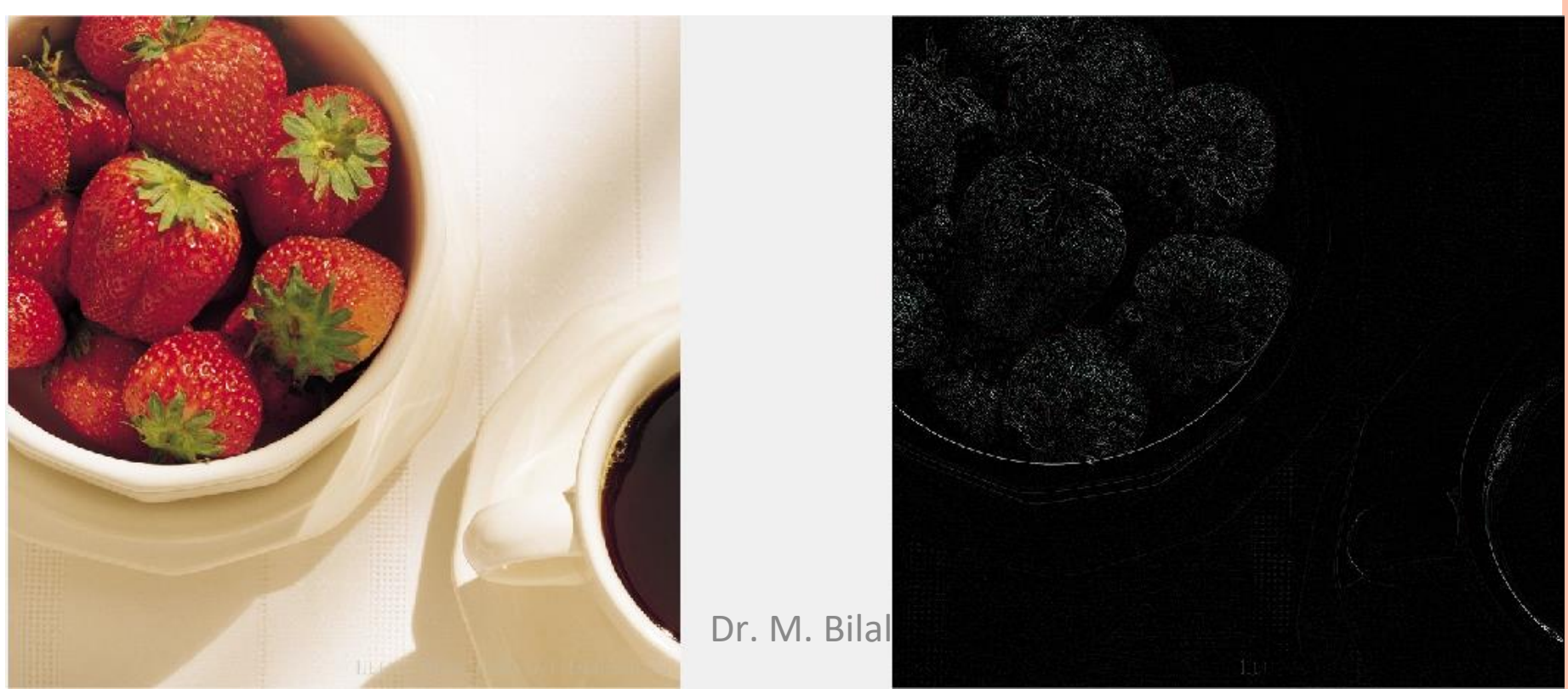
- Thus we conclude that smoothing by neighborhood averaging can be carried out on a per color plane basis.

Color Image Smoothing



Color Image Sharpening

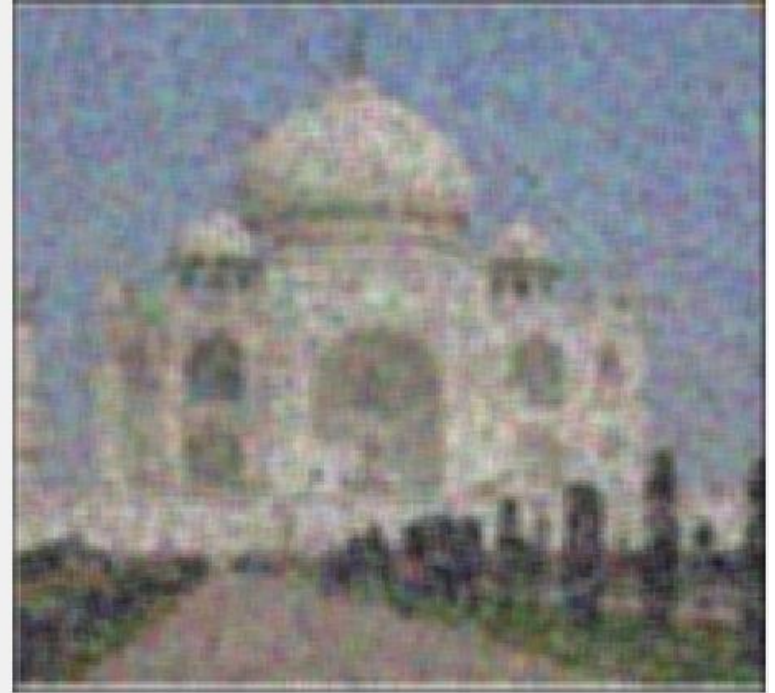
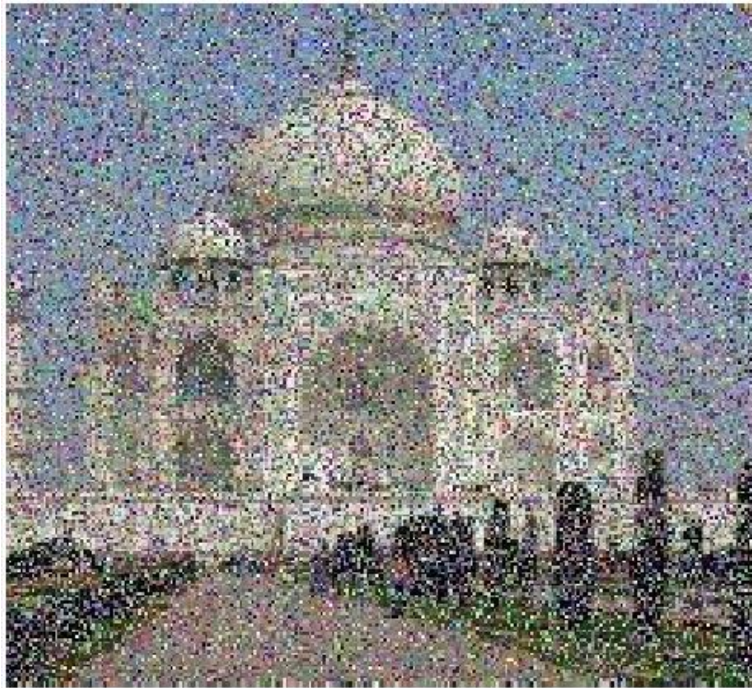
- Using Laplacian filter apply on each R, G and B channel to get sharp image



NOISE IN COLOR IMAGES

- Noise in color images can be removed through various noise models which we use in Image Restoration in case the noise content of a color image has the same characteristics in each color channel.
- But it is possible for color channels to be affected differently by noise so in this case noise are removed from the image by independently processing each plane
- Remove noise by applying smoothing filters (e.g gaussian, average, median) to each plane individually and then combine the result.

NOISE IN COLOR IMAGES



COLOR IMAGE COMPRESSION

- Compression is the process of reducing or eliminating redundant and/or irrelevant information
- A compressed image is not directly displayable it must be decompressed before input to a color monitor.
- In case if in a compressed image 1 bit of data represents 230 bits of data in the original image, then compressed image could be transmitted over internet in 1 minute as compared to original image which will take 4 hours to transmit.



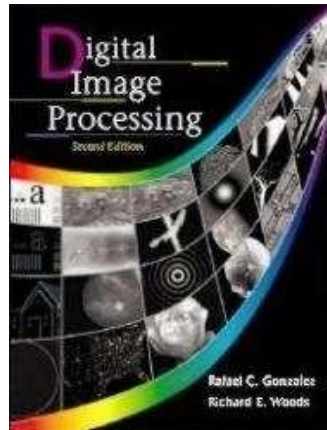
a b
c d

FIGURE 6.51

Color image compression.
(a) Original RGB image. (b) Result of compressing and decompressing the image in (a).

REFERENCES

- “Digital Image Processing”, 2/ E, Rafael C. Gonzalez & Richard E. Woods, www.prenhall.com/gonzalezwoods.



Any question



THE END

